kiosk-admin

RDP Browser-Kiosk Server with Web Management Console
Installation & User Guide

Version 1.3.0 · July 2026

1. What is kiosk-admin?

kiosk-admin turns a single Ubuntu/Debian machine into a multi-user **RDP kiosk server**. Users connect with any standard RDP client (Windows *Remote Desktop*, Remmina, FreeRDP, mobile RDP apps) and receive a **locked-down browser with a fixed set of tabs — 5 by default**. The tab strip at the top of the screen is the **only** visible browser UI: no address bar, no menus, no settings, no developer tools, no downloads. The default tabs are permanent (no close button); links that open in a new tab appear *after* them with a normal close button, so users can dismiss those when done. If the browser crashes or is closed, it restarts by itself within seconds.

Administrators manage everything from a web console:

- **Live dashboard** — every logged-in user with an auto-updating screenshot of their screen
- **Screen mirroring & remote control** — click any session to watch it live or take over mouse and keyboard (noVNC), with a “view only” toggle
- **Hard refresh** — send Ctrl+Shift+R to a user's browser remotely
- **Browser reset** — kill and relaunch a user's browser with a clean profile (clears cookies, history, and any wedged state)
- **User management** — create kiosk users, set passwords, delete users
- **Session logout** — terminate any user's RDP session

Architecture at a glance

```
RDP clients ──> xrdp ──> per-user Xorg session (:10, :11, ...)
                              ├──> Firefox, tab-strip-only UI  (respawn loop = self-healing)
                              └──> x11vnc on 127.0.0.1:59<display>  (mirroring)

Admin browser ──> kiosk-admin web console (HTTPS, port 8443)
                     ├── session list, screenshots, user management
                     └── websocket bridge ──> x11vnc ──> live view & control
```

Security model: the web app runs as an unprivileged user; every privileged action goes through a single audited root helper (`kiosk-ctl`) allowed by one sudoers line. VNC traffic never leaves localhost — the only network entry points are RDP (3389) and the password-protected admin console (8443, TLS).

2. Requirements

Item	Requirement
Operating system	Ubuntu 22.04 / 24.04 or Debian 12 (server or desktop)
Access	Root (sudo) on the target machine
Network	Internet access during install (apt packages); ports 3389 and 8443 reachable by clients
Hardware	Roughly 400–600 MB RAM per concurrent kiosk user (Firefox with 5 tabs); 2 GB + 600 MB/user is a safe sizing rule

The installer pulls everything else automatically: xrdp, xorgxrdp, openbox, x11vnc, xdotool, ImageMagick, noVNC, Firefox, and a private Python environment.

3. Installation

1. Copy this repository to the target machine (git clone, scp, or unpack the release tarball):

   ```
   git clone <repo-url> kiosk-admin
   cd kiosk-admin
   ```

2. Run the installer:

   ```
   make install
   ```

   (equivalent to `sudo bash deploy/install.sh`)
3. At the end, the installer prints the console address and a **generated admin password**:

   ```
   == done: kiosk-admin 1.0.0 ==
      admin console : https://192.168.1.50:8443
      admin password: Xy9kQ2mP4LcA   (change with: sudo kiosk-admin-passwd)
   ```

> **Save the admin password now.** It is shown only once. You can replace it at any time with `sudo kiosk-admin-passwd`.

3.1 First-time setup

1. **Set the kiosk tabs.** Edit `/etc/kiosk/kiosk.conf` — one URL per tab, space-separated, opened left to right (5 by default):

   ```
   KIOSK_URLS="https://erp.example.com https://mail.example.com https://wiki.example.com https://crm.example.com
   https://reports.example.com"
   ```

   The change applies the next time each user's browser starts — use the console's **Reset** button to apply it to running sessions immediately.
2. **Open the firewall** (if enabled) for the two service ports only:

   ```
   sudo ufw allow 3389/tcp    # RDP for users
   sudo ufw allow 8443/tcp    # admin console
   ```

3. **Log in to the console** at `https://<server-ip>:8443`. The TLS certificate is self-signed, so your browser shows a one-time warning — accept it (or install real certificates, see §7).
4. **Create the first user** with the **+ Add user** button (username: lowercase letters, digits, `-`, `_`; password: minimum 6 characters).
5. **Test the user login:** from any PC, open an RDP client, connect to the server's address, and sign in with the new username and password. The browser should fill the screen with the configured tabs across the top and no other UI — and a new session card should show up on the admin dashboard within a few seconds.

4. Using the admin console

4.1 Dashboard

The main page shows one card per active RDP session with the username, display number, session uptime, a green/red dot for browser health, and a screenshot that refreshes automatically every few seconds.

Button	What it does	When to use it
Refresh	Sends Ctrl+Shift+R to the user's browser (cache-bypassing page reload)	The page shows stale data or a deployment just went out
Reset	Kills the browser; it relaunches within ~2 seconds with a brand-new profile (cookies, history, and session state wiped; saved website passwords are kept) and the full default tab set restored	The browser is frozen, logged into the wrong account, tabs were lost, or the tab list in `kiosk.conf` changed
Logout	Terminates the user's whole RDP session	Freeing a seat, or ending an abandoned session

4.2 Screen mirroring & remote control

1. Click a session's screenshot (or open it in a new tab). The live viewer connects in about a second.
2. By default **you control the session**: your mouse and keyboard act on the user's screen — ideal for guided support.

3. Tick **view only** in the toolbar to watch without interfering.
4. The viewer toolbar also has **Refresh page** and **Reset browser** buttons for the session being viewed.

> The user keeps full control of their own session at all times — mirroring is shared, not exclusive. Both sides see the same screen.

4.3 Managing users

- **Add:** “+ Add user” in the header. The account works for RDP immediately.
- **Set password:** per-user button in the *Kiosk users* table.
- **Delete:** per-user button; logs the user out first, then removes the account **and its home directory**. This cannot be undone.
- The table also shows who is currently online.

> Only accounts in the `kioskusers` group are visible to (and manageable by) the console. Regular admin accounts on the same machine are untouched and still get a normal desktop over RDP.

5. Upgrading

Upgrading **is** reinstalling — the installer is idempotent:

```
cd kiosk-admin
git pull                  # or unpack the new release tarball over this directory
make upgrade              # same as make install
```

Survives every upgrade	Replaced by the upgrade
`/etc/kiosk/kiosk.conf` (kiosk URL) Admin password, cookie secret, TLS certs All kiosk user accounts and home dirs **Live RDP sessions** (xrdp is never restarted)	Web app in `/opt/kiosk-admin/` Scripts in `/usr/local/bin/` `/etc/xrdp/startwm.sh`, Chromium policies systemd unit, Python dependencies

Only the admin console restarts (1–2 seconds); users notice nothing. Updated session scripts take effect at each user's next login. To upgrade an offline server: on any machine run `make package`, copy the resulting `kiosk-admin-<version>.tar.gz` across, unpack, `make install`.

6. Uninstalling

```
make uninstall    # removes app + scripts, KEEPS configs (easy reinstall later)
make purge        # also deletes /etc/kiosk and /etc/kiosk-admin
```

Kiosk user accounts are never deleted automatically — remove them from the web UI beforehand, or afterwards with `sudo userdel -r <name>`. xrdp itself stays installed; the original `startwm.sh` is restored.

7. Configuration reference

/etc/kiosk/kiosk.conf

Setting	Default	Meaning
`KIOSK_URLS`	5 example sites	The user's tabs: one URL per tab, space-separated, opened left to right. The tab strip is the only browser UI users see.
`KIOSK_FRESH_PROFILE`	yes	`yes` : wipe the browser profile on every (re)start — no history, cookies, or crash-restore prompts. `no` : keep state.
`KIOSK_SAVE_LOGINS`	yes	`yes` : Firefox offers to save website passwords, and saved logins survive profile wipes, browser restarts, and admin Resets. `no` : password saving is disabled and existing saved logins are wiped on the next restart.
`KIOSK_RESTART_DELAY`	2	Seconds before relaunching a closed/crashed browser
`KIOSK_BG`	#1a1a2e	Desktop color visible briefly before the browser starts

Other locations

Path	Purpose
`/etc/kiosk-admin/admin.passwd`	Console password (bcrypt); change via `sudo kiosk-admin-passwd`
`/etc/kiosk-admin/certs/`	TLS cert/key. Replace `cert.pem` / `key.pem` with real ones, then `sudo systemctl restart kiosk-admin`
`/opt/kiosk-admin/`	Installed web app (replaced on upgrade — never edit here)
`~<user>/.kiosk/`	Per-user logs: browser.log, x11vnc.log, session.log

Ports

Port	Service	Exposure
3389/tcp	xrdp — user logins	Open to users
8443/tcp	Admin console (HTTPS + WebSocket)	Open to admins only, ideally restricted by firewall source IP
59xx/tcp	x11vnc per session	127.0.0.1 only — never exposed

8. Troubleshooting

Symptom	Cause & fix
Console unreachable on :8443	Check `systemctl status kiosk-admin` and `journalctl -u kiosk-admin -e`; confirm the firewall allows 8443.
User gets a black/empty RDP screen	Browser may be starting; check `~<user>/.kiosk/browser.log`. Verify Firefox is installed and the `KIOSK_URLS` entries are valid.
“no active session” on Refresh/Screenshot	The user has no Xorg process — they disconnected and the session was reaped. They simply need to reconnect.
Live viewer says “connection lost”	Your console login expired (sign in again on the dashboard) or the session ended. x11vnc log: `~<user>/.kiosk/x11vnc.log`.
Duplicate sessions for one user	Reconnects with a different color depth create a second session. Set `Policy=Default` and `KillDisconnected=false` in `/etc/xrdp/sesman.ini`, restart xrdp off-hours.
Wrong admin password / lost password	`sudo kiosk-admin-passwd` — effective immediately.
Tab list change not visible	`KIOSK_URLS` is read at browser start. Press **Reset** on each session (or wait for the users' next login).
A user closed some of their default tabs	Default tabs have no close button, but Ctrl+W still works. Press **Reset** to restore the full default tab set. (Tabs opened from links are meant to be closeable — that is by design.)